Meeting Record

Mid point circle algo:

if $m < 1$

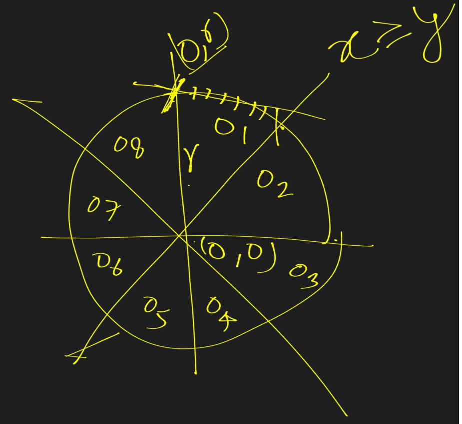
$x_{k+1} = (x_k + 1)$

either $y_{k+1} = y_k$ or $y_k - 1$

∴ next coordinate:

either $(x_k + 1, y_k)$ or

$(x_k + 1, y_k - 1)$



Calculate mid point

$$= \left(\frac{x_k + 1 + x_k + 1}{2}, \frac{y_k + y_k - 1}{2} \right)$$

$$\boxed{x^2 + y^2 = r^2}$$

$$= \left(x_k + 1, \; y_k - \tfrac{1}{2} \right)$$

$P_k = x^2 + y^2 - r^2$

$P_k = (x_k + 1)^2 + (y_k - \tfrac{1}{2})^2 - r^2$ ——— (1)

$$P_{k+1} = (x_{k+1}+1)^2 + \left(y_{k+1} - \tfrac{1}{2}\right)^2 - r^2$$

$$= ((x_k+1)+1)^2 + \left(y_{k+1} - \tfrac{1}{2}\right)^2 - r^2$$

$$P_{k+1} = (x_k+1)^2 + 2(x_k+1) + 1 + y_{k+1}^2 - y_{k+1} + \tfrac{1}{4} - r^2$$

$$P_{k+1} - P_k = \cancel{(x_k+1)^2} + 2(x_k+1) + 1 + y_{k+1}^2 - y_{k+1} + \tfrac{1}{4} - \cancel{r^2}$$

$$- \cancel{(x_k+1)^2} - \left[y_k^2 - y_k + \tfrac{1}{4}\right] + \cancel{r^2}$$

$$P_{k+1} - P_k = 2(x_k+1) + 1 + y_{k+1}^2 - y_{k+1} + \cancel{\tfrac{1}{4}} - y_k^2 + y_k - \cancel{\tfrac{1}{4}}$$

$$P_{k+1} = P_k + 2(x_k+1) + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1$$

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initial point $(0, r)$

$$P_k = (x_k+1)^2 + \left(y_k - \tfrac{1}{2}\right)^2 - r^2$$

$$= (0+1)^2 + \left(r - \tfrac{1}{2}\right)^2 - r^2$$

$$1 + \cancel{r^2} - r + \tfrac{1}{4} - \cancel{r^2}$$

$$1 + \tfrac{1}{4} - r$$

$$\tfrac{5}{4} - r \approx 1 - r$$

$$\therefore \boxed{P_0 = 1 - r}$$

if $P_k \geq 0$	if $P_k < 0$
$x_{k+1} = x_k + 1$	$x_{k+1} = x_k + 1$
$y_{k+1} = y_k - 1$	$y_{k+1} = y_k$

Chapter 3: 2D Geometric transformation:

Basic transformations

(i) Translation
(ii) Scaling
(iii) Rotation

Translation: Moving object from one place to another place.
i.e. changing the position.

Scaling: To resize the object either maximize or minimize.

Rotation: Rotate an object using certain angle.

$(0,1)\ (5,6)$

$$m = \frac{y_{k+1} - y_k}{x_{k+1} - x_k} = \frac{6-1}{5-0} = \frac{5}{5} = 1$$

$m = 1$, both x & y moves in unit

$x_{k+1} = x_k + 1$

$y_{k+1} = y_k + 1$

Steps	x_k	y_k	x_{k+1}	y_{k+1}
1	0	1	1	2
2	1	2	2	3
3	2	3	3	4
4	3	4	4	5
5	4	5	5	6